

Lecture Seventeen

Topic for today: Estimation in AR models.
AR(1)

Revisit: Linear Regression

$$(x_1, y_1), \dots, (x_n, y_n)$$

$$(A) \rightarrow y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$
$$\theta = (\beta_0, \beta_1, \sigma)$$

$$\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

(MLE) likelihood: $\prod_{i=1}^n \left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \right)$

Derivation
of the likelihood

density of observations (assuming parameters are fixed).

$$f(x_1, y_1, x_2, y_2, \dots, x_n, y_n)$$

$$x_1, y_1, x_2, y_2, \dots, x_n, y_n \mid \theta$$

$$= f(y_1, \dots, y_n) f(x_1, \dots, x_n)$$
$$y_1, \dots, y_n \mid x_1, \dots, x_n, \theta \quad x_1, \dots, x_n \mid \theta$$

$$= f(y_1, \dots, y_n)$$
$$\beta_0 + \beta_1 x_1 + \varepsilon_1, \beta_0 + \beta_1 x_2 + \varepsilon_2, \dots, \beta_0 + \beta_1 x_n + \varepsilon_n \mid x_1, \dots, x_n, \theta$$

$$= f(y_1 - \beta_0 - \beta_1 x_1, y_2 - \beta_0 - \beta_1 x_2, \dots, y_n - \beta_0 - \beta_1 x_n) \times f(x_1, \dots, x_n)$$
$$\varepsilon_1, \dots, \varepsilon_n \mid x_1, \dots, x_n, \theta \quad x_1, \dots, x_n \mid \theta$$

$$f(y)$$
$$\beta_0 + \beta_1 x + \varepsilon$$
$$= f(y - \beta_0 - \beta_1 x)$$
$$\varepsilon$$

Assume: $\varepsilon_1, \dots, \varepsilon_n$ are independent of $x_1, \dots, x_n$ given θ .

A2

$$= f(y_1 - \beta_0 - \beta_1 x_1, \dots, y_n - \beta_0 - \beta_1 x_n) f_{x_1, \dots, x_n | \theta}$$

A3: $\varepsilon_1, \dots, \varepsilon_n \stackrel{iid}{\sim} N(0, \sigma^2)$

$$= \left[\prod_{i=1}^n f_{\varepsilon_i | \theta}(y_i - \beta_0 - \beta_1 x_i) \right] \times f_{x_1, \dots, x_n | \theta}$$

$$= \left[\prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \right] f_{x_1, \dots, x_n | \theta}$$

A4: $f_{x_1, \dots, x_n | \theta}$ does not depend on θ .

$$\propto \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right)$$

① $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, i = 1, \dots, n$

② $(\varepsilon_1, \dots, \varepsilon_n)$ is independent of $(x_1, \dots, x_n)$

③ $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

④ density of $x_1, \dots, x_n$ does not depend on $(\beta_0, \beta_1, \sigma)$

likelihood: $\prod_{i=1}^n \left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \right]$

Maximize likelihood to get MLEs.

$\hat{\beta}_0, \hat{\beta}_1$: least squares estimates

$$\hat{\sigma}_{MLE}^2 = \frac{RSS}{n}$$

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

Bayesian: $\beta_0, \beta_1, \log \sigma$ iid $Unif(-C, C)$

$$\beta \mid \text{data} \sim t_{n-2, 2} \left(\hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1} \right)$$

$$\hat{\sigma}^2 = \frac{RSS}{n-2}$$

Auto Regression AR(1)

(A) $\rightarrow y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, t=2, \dots, n$

$y_t = \phi_0 + \phi_1 x_t + \varepsilon_t, t=2, \dots, n$
with $x_t = y_{t-1}$

$\theta = \begin{pmatrix} \phi_0 \\ \phi_1 \end{pmatrix}$

How to write the likelihood now?

$$f(y_1, \dots, y_n)$$

$$f(y_1, \dots, y_n \mid \theta)$$

$$= f(y_1 \mid \theta) f(y_2 \mid y_1, \theta) f(y_3 \mid y_1, y_2, \theta) \dots f(y_n \mid y_1, \dots, y_{n-1}, \theta)$$

$$= f(y_1 \mid \theta) \prod_{t=2}^n f(y_t \mid y_1, \dots, y_{t-1}, \theta)$$

$$= f_{y_1|\theta} \prod_{t=2}^n f(y_t | \phi_0 + \phi_1 y_{t-1} + \varepsilon_t | y_1 \dots y_{t-1}, \theta)$$

$$= f_{y_1|\theta} \prod_{t=2}^n f(\varepsilon_t | y_t - \phi_0 - \phi_1 y_{t-1} | y_1 \dots y_{t-1}, \theta)$$

(A2) ε_t is independent of $y_1 \dots y_{t-1}$ (given θ)
for each $t = 2, \dots, n$

$$= f_{y_1|\theta} \prod_{t=2}^n f(\varepsilon_t | y_t - \phi_0 - \phi_1 y_{t-1})$$

(A3) $\varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$

$$= f_{y_1|\theta} \left[\prod_{t=2}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (y_t - \phi_0 - \phi_1 y_{t-1})^2\right) \right]$$

$$\left[\prod_{t=2}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (y_t - \phi_0 - \phi_1 x_t)^2\right) \right] \quad x_t = y_{t-1}$$

Same likelihood as in linear regression.

(A1) $y_t = \phi_0 + \phi_1 x_t + \varepsilon_t$ with $x_t = y_{t-1}$

(A2) ε_t is independent of $y_1 \dots y_{t-1}$, $t = 2, \dots, n$

(A3) $\varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$

↓
A4 The density of y , does not depend on θ .

Under $A1, A2, A3, A4$,

$$\text{likelihood} \propto \prod_{t=2}^{n-1} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2\sigma^2} (y_t - \phi_0 - \phi_1 y_{t-1})^2\right)$$

identical to linear regression likelihood.

MLE: same as in linear regression

$$\hat{\phi} = (X^T X)^{-1} X^T y \quad X = \begin{bmatrix} 1 & y_1 \\ \vdots & \vdots \\ 1 & y_{n-1} \end{bmatrix}$$

$$y = \begin{pmatrix} y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$\hat{\sigma}_{MLE} = \sqrt{\frac{RSS}{n-1}}, \quad \hat{\sigma} = \sqrt{\frac{RSS}{(n-1)-2}}$$

$\phi_0, \phi_1, \log \sigma \stackrel{iid}{\sim} \text{Unif}(-C, C)$

$$\phi \mid \text{data} \sim t_{n-3, 2} \left(\hat{\phi}, \hat{\sigma}^2 (X^T X)^{-1} \right)$$

↑
least squares

$\hat{\phi} = (X^T X)^{-1} X^T y$ can longer say:

$$\hat{\phi} \sim N(\phi, \sigma^2 (X^T X)^{-1})$$

standard errors corresponding to ϕ_0, ϕ_1 :

$$\rightarrow \sqrt{\text{diag}[\hat{\sigma}^2 (X^T X)^{-1}]}$$

$$\hat{\sigma} = \sqrt{\frac{RSS}{\text{residual degrees of freedom}}}$$

AutoReg: $\sqrt{\text{diag}(\hat{\sigma}_{MLE}^2 (X^T X)^{-1})}$ $\hat{\sigma}_{MLE}^2 = \frac{RSS}{n-1}$
 AR(1)

How to write a model for $y_t | \theta$?

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, t=2, \dots, n$$

$$y_1 = \phi_0 + \phi_1 y_0 + \varepsilon_1$$

$$= \phi_0 + \phi_1 (\phi_0 + \phi_1 y_{-1} + \varepsilon_0) + \varepsilon_1$$

$$= \phi_0 + \phi_1 \phi_0 + \varepsilon_1 + \phi_1 \varepsilon_0 + \phi_1^2 y_{-1}$$

$$= \phi_0 (1 + \phi_1) + \varepsilon_1 + \phi_1 \varepsilon_0 + \phi_1^2 y_{-1}$$

$$= \phi_0 (1 + \phi_1 + \phi_1^2) + \varepsilon_1 + \phi_1 \varepsilon_0 + \phi_1^2 \varepsilon_{-1} + \phi_1^3 y_{-2}$$

$$= \phi_0 (1 + \phi_1 + \phi_1^2 + \dots + \phi_1^M) + \varepsilon_1 + \phi_1 \varepsilon_0 + \dots + \phi_1^M \varepsilon_{1-M}$$

$$+ \phi_1^{M+1} y_{-M}$$

Suppose $|\phi_1| < 1$.

Let $M \rightarrow \infty$ and assume $|\phi_1| < 1$:

$$y_1 = \phi_0 (1 + \phi_1 + \phi_1^2 + \dots) + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{1-j}$$

$$y_1 = \frac{\phi_0}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{1-j}$$

$$y_1 \sim N\left(\frac{\phi_0}{1-\phi_1}, \sum_{j=0}^{\infty} \phi_1^{2j} \sigma^2\right)$$

$$= N\left(\frac{\phi_0}{1-\phi_1}, \frac{\sigma^2}{1-\phi_1^2}\right), (|\phi_1| < 1)$$

Likelihood

$$= \frac{\sqrt{1-\phi_1^2}}{\sqrt{2\pi} \sigma} \exp\left[-\frac{\left(y_1 - \frac{\phi_0}{1-\phi_1}\right)^2 (1-\phi_1^2)}{2\sigma^2}\right]$$

$$\left[\prod_{t=2}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left[-\frac{(y_t - \phi_0 - \phi_1 y_{t-1})^2}{2\sigma^2}\right] \right]$$

$$(|\phi_1| < 1)$$

Two AR(1) Models

- ① y_1 has a density not depending on parameters
(y_1 treated as constant)
- ② $y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, t=2, \dots, n$
- ③ ε_t independent of $y_1, \dots, y_{t-1}$ ←
- ④ $\varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$

↓
FIRST AR(1)

Second AR(1)

$$① \quad y_1 = \frac{\phi_0}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{1-j}$$

$$② \quad y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, \quad t=2, \dots, n.$$

$$\begin{aligned} y_2 &= \phi_0 + \phi_1 y_1 + \varepsilon_2 \\ &= \phi_0 + \phi_1 \left[\frac{\phi_0}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{1-j} \right] + \varepsilon_2 \\ &= \phi_0 + \frac{\phi_1 \phi_0}{1 - \phi_1} + \left[\sum_{j=0}^{\infty} \phi_1^{j+1} \varepsilon_{1-j} + \varepsilon_2 \right] \\ &\quad \varepsilon_2 + \phi_1 \varepsilon_1 + \phi_2 \varepsilon_0 + \dots \\ &= \frac{\phi_0}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{2-j} \end{aligned}$$

Second AR(1)

$$|\phi_1| < 1$$

$$① \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma^2),$$

$$② \quad y_t = \frac{\phi_0}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \varepsilon_{t-j} \quad \text{for every } t$$

y_t is calculated from $\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots$

for every $t = \dots, -3, -2, -1, 0, 1, 2, \dots$

STATIONARY

$$\varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

$$\varepsilon_t + \frac{1}{2} \varepsilon_{t-1} + \frac{1}{4} \varepsilon_{t-2} + \frac{1}{8} \varepsilon_{t-3} + \dots$$